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1. The relation between machine learning and PDE

- A Machine Learning Approach for Solving Stochastic PDEs
- <https://arxiv.org/abs/2509.00204> (<https://arxiv.org/abs/2509.00204>).
- <https://doi.org/10.48550/arXiv.2509.00204> (<https://doi.org/10.48550/arXiv.2509.00204>).

2. How exponential forms of other distributions are like

- Exponential Families and Their Applications in Statistical Learning
- <https://arxiv.org/abs/1009.2754> (<https://arxiv.org/abs/1009.2754>).
- The Use of Exponential and Gamma Distributions for Modeling System Reliability
- <https://www.jstor.org/stable/2284394> (<https://www.jstor.org/stable/2284394>).

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簡化疾病模型

定義一個可控、可模擬的疾病模型

- 病人特徵

每位病人有3個參數

$$x = (x_1, x_2, x_3) \sim \mathcal{N}(0, I)$$

- 藥物劑量

$$a \in 0, 1, 2 \quad (\text{無治療, 中劑量, 高劑量})$$

- 治療效果

$$\text{efficacy} = 0.8a - 0.3a^2 + 0.5x_1 - 0.1x_2 + \epsilon_1$$

- 副作用

$$\text{side_effect_prob} = \sigma(0.2a^2 + 0.3x_3)$$

- 最終結果

最大化治療效果並最小化副作用

$$R = \text{efficacy} - 2 \times \text{side_effect_prob}$$

第一步：生成簡化虛擬臨床資料

```
import numpy as np

# generate virtual patients
N = 5000
X = np.random.normal(0, 1, (N, 3))
A = np.random.choice([0,1,2], N)

# true efficacy model
efficacy = 0.8*A - 0.3*(A**2) + 0.5*X[:,0] - 0.1*X[:,1]
+ np.random.normal(0, 0.1, N)

# true side effect probability
side_prob = 1 / (1 + np.exp(-(0.2*(A**2) + 0.3*X[:,2])))
side_effect = np.random.binomial(1, side_prob)

# dataset
data = {
    "X": X,
    "A": A,
    "efficacy": efficacy,
    "side_effect": side_effect
}

print("Data example:", X[0], A[0], efficacy[0], side_effect[0])
```

第二步：訓練一個模型預測功效與副作用

```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor, RandomForestClassifier

# combine features and dose
features = np.column_stack([X, A])

# split
X_train, X_test, y_eff_train, y_eff_test
= train_test_split(features, efficacy, test_size=0.2)
_, _, y_se_train, y_se_test
= train_test_split(features, side_effect, test_size=0.2)

# efficacy prediction model
eff_model = RandomForestRegressor()
eff_model.fit(X_train, y_eff_train)

# side effect prediction model
se_model = RandomForestClassifier()
se_model.fit(X_train, y_se_train)

print("Efficacy R^2:", eff_model.score(X_test, y_eff_test))
print("Side effect acc:", se_model.score(X_test, y_se_test))
```

第三步：強化學習挑選最佳劑量

環境：

- state = biomarkers
- action = {0,1,2}
- reward = efficacy - 2 × side effect probability

```

Q = np.zeros(3)
counts = np.zeros(3)

epsilon = 0.1

for episode in range(5000):
    # sample a new virtual patient
    x = np.random.normal(0,1,3)

    # choose action
    if np.random.rand() < epsilon:
        a = np.random.choice([0,1,2])
    else:
        a = np.argmax(Q)

    # predict efficacy and side effect probability
    features = np.append(x, a).reshape(1, -1)
    eff = eff_model.predict(features)[0]
    se_prob = se_model.predict_proba(features)[0][1]

    reward = eff - 2*se_prob

    # update Q
    counts[a] += 1
    lr = 1 / counts[a]
    Q[a] = Q[a] + lr*(reward - Q[a])

print("Learned Q-values:", Q)
print("Optimal dose learned:", np.argmax(Q))

```

展示的未來能力雛型

1. 虛擬病人生成

雖然只有3個參數，但概念一致，可以模擬群體後用於虛擬試驗。

2. 模擬藥物作用

雖然是真實函數，但已經代表未來數據模擬的模型。

3. 效果與副作用的機器學習預測

對應於未來的：

- survival rate
- symptom reduction
- side effects

4. 強化學習治療策略探索

是由虛擬臨床試驗往最佳療法搜尋推進的最小版本。